

SEA SENTRY

(FISHERMAN BORDER ALERT SYSTEM)

(PHASE-2)

TEAMMATES:

- AASHIKA M
- INIYA M
- DEEPIKA G

BRAINSTORMING REPORT:

Problems Faced by Tamil Nadu Fishermen

Tamil Nadu fishermen depend on the sea for their livelihood, but fishing near **Katchatheevu** has become increasingly challenging. Due to **climate change, changing sea conditions, and reduced fish availability**, fishermen often travel farther into the sea in search of productive fishing grounds. Since the **International Maritime Boundary Line (IMBL)** has no physical markers, fishermen cannot easily identify the exact boundary while fishing and may unknowingly enter restricted waters.

Crossing the maritime boundary can lead to the **detention of fishermen, seizure of fishing boats and equipment, loss of income, and severe hardship for their families**. Long hours at sea, harsh weather conditions, and the pressure to find fish further increase the risk of accidental boundary crossings. These incidents affect not only individual fishermen but also the coastal communities that depend on fishing as their primary source of livelihood.

Therefore, there is a need for a **simple, low-cost, and easy-to-use warning system** that alerts fishermen before they approach the maritime boundary. A **GPS-based system with an LED indicator and buzzer** can provide an immediate visual and audible warning, helping fishermen remain within safe fishing zones and reducing accidental crossings near **Katchatheevu**.

Brainstorming Ideas

Idea 1: GPS-Based Maritime Boundary Alert System

- Uses **ESP32**, **Neo-6M GPS**, **LED**, and **Buzzer**.
- Warns fishermen before they reach the International Maritime Boundary Line (IMBL).
- Simple, affordable, and works without internet.

Idea 2: Weather Alert System

- Monitors weather conditions and warns fishermen about storms and rough seas.
- Requires weather data, making implementation more complex.

Idea 3: Fishermen Health Monitoring System

- Monitors the fisherman's heart rate and body temperature.
- Sends alerts during emergencies.
- More suitable for healthcare than maritime safety.

Idea 4: Water Pollution Monitoring System

- Measures water quality using sensors.
- Helps identify polluted fishing areas.
- Does not solve the maritime boundary problem.

Idea 5: Smart Fish Detection System

- Uses sensors to identify areas with higher fish availability.
- Reduces search time for fishermen.
- Requires additional hardware and has a higher cost.

SELECTED IDEA:

Our team has selected to develop **Sea Sentry**, an **ESP32-based Maritime Boundary Alert System** for Tamil Nadu fishermen. The system uses a **Neo-6M GPS module** to continuously monitor the boat's location. When the boat approaches or crosses the predefined **International Maritime Boundary Line (IMBL)** near **Katchatheevu**, the **ESP32** processes the GPS coordinates and immediately activates a **red LED** and a **buzzer** to provide both visual and audible warnings. This helps fishermen recognize the danger and change their course before entering restricted waters.

The proposed solution is **simple, affordable, reliable, and easy to operate**. It does not require an internet connection or Wi-Fi, making it suitable for small fishing boats operating in remote sea areas. By providing timely alerts, the system aims to reduce accidental maritime boundary crossings, improve fishermen's safety, and protect their livelihood.

Reason for Selection

Our team selected this idea because **accidental crossing of the International Maritime Boundary Line (IMBL) near Katchatheevu is a major challenge faced by Tamil Nadu fishermen**. Since there are no visible boundary markers in the sea, fishermen often unknowingly enter restricted waters while searching for fish, which can result in detention, confiscation of boats, and financial losses.

The proposed **ESP32-based GPS Maritime Boundary Alert System** provides a **simple, low-cost, and reliable** solution by warning fishermen through an **LED and buzzer** before they approach the maritime boundary. The system operates **without requiring an internet connection**, making it practical for use in remote sea areas. This idea was selected because it directly addresses a real-world problem, enhances fishermen's safety, protects their livelihood, and is feasible to develop using readily available components within the project's scope.

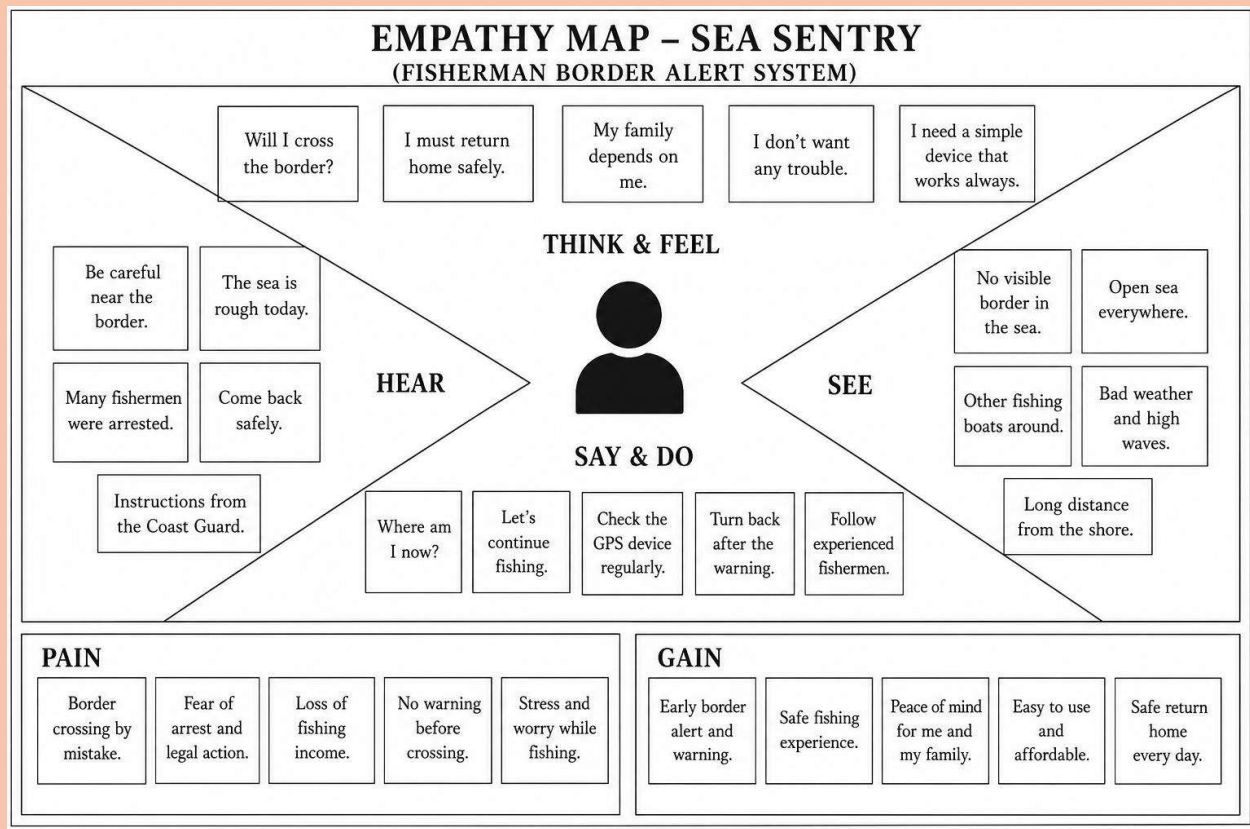
Idea Prioritization Matrix

Proposed Idea	Feasibility (5)	Cost (5)	Impact (5)	Innovation (5)	Ease of Implementation (5)	Total (25)	Priority
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GPS-
Based
Maritime
Boundary
Alert
(ESP32 +
GPS +
LED +
Buzzer)
Water
Pollution
Monitorin
g System
Fisherme
n Health
Monitorin
g Device
Smart
Fish
Detection
System
Weather
Alert
System

5	4	5	4	5	23	1
3	3	4	4	3	17	2
3	2	4	3	2	14	3
2	2	5	4	2	15	4
3	3	4	3	3	16	5

EMPATHY MAP:



Proposed Solution

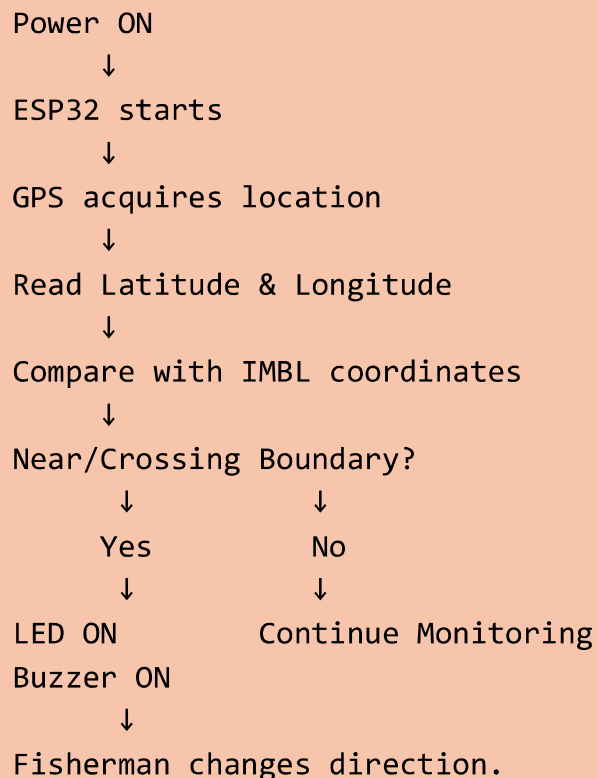
The proposed solution is **Sea Sentry**, an **ESP32-based Maritime Boundary Alert System** designed to improve the safety of Tamil Nadu fishermen operating near **Katchatheevu**. The system uses a **Neo-6M GPS module** to continuously monitor the boat's geographic location. The coordinates are compared with predefined International Maritime Boundary Line (IMBL) coordinates stored in the ESP32.

When the boat approaches or crosses the predefined boundary, the **ESP32 immediately activates an LED and a buzzer** to provide clear visual and audible warnings. These alerts notify the fishermen to change their direction and remain within permitted fishing waters.

Since the system operates using GPS, it **does not require an internet connection or Wi-Fi**, making it suitable for use in remote offshore areas.

The proposed solution is **simple, reliable, cost-effective, and easy to use**. It can be installed on small fishing boats and helps reduce accidental maritime boundary crossings, thereby improving fishermen's safety, protecting their livelihood, and minimizing incidents of boat seizure and detention.

Flow of the Proposed Solution



Functional Requirements

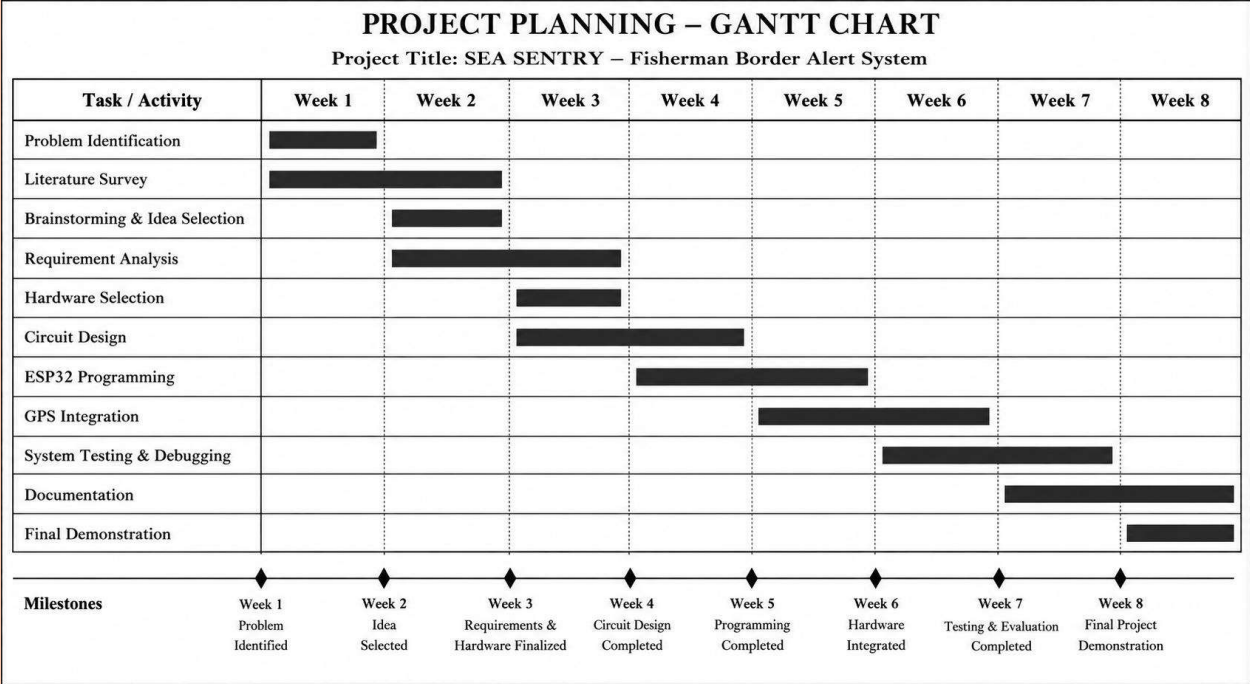
1. The system shall continuously receive the boat's location using the **Neo-6M GPS module**.
2. The system shall compare the current GPS coordinates with the predefined **International Maritime Boundary Line (IMBL)** coordinates.
3. The system shall detect when the boat approaches or crosses the predefined maritime boundary.

4. The system shall activate the **LED** to provide a visual warning when the boat is near or crosses the boundary.
5. The system shall activate the **buzzer** to provide an audible warning when the boat is near or crosses the boundary.
6. The system shall continuously monitor the boat's location while the device is powered on.
7. The system shall stop the warning once the boat returns to the safe fishing zone.
8. The ESP32 shall control all system operations and process the GPS data.

Non-Functional Requirements

1. The system should provide **accurate GPS-based boundary detection**.
2. The system should operate **without an internet or Wi-Fi connection**.
3. The system should be **easy to install and operate** on small fishing boats.
4. The system should have **low power consumption** for longer operation.
5. The system should provide **quick warning alerts** with minimal delay.
6. The system should be **reliable** under normal marine operating conditions.
7. The system should be **cost-effective** by using readily available electronic components.
8. The system should be **compact and lightweight** so that it can be easily mounted on a fishing boat.

PROJECT PLANNING (GANTT CHART):



RISK ANALYSIS:

Risk ID	Potential Risk	Impact	Probability	Mitigation Strategy
R1	Weak or no GPS signal	High	Medium	Wait for a valid GPS fix before monitoring and test in open areas.
R2	Incorrect IMBL coordinates	High	Low	Verify boundary coordinates from reliable sources before programming.
R3	ESP32 hardware failure	High	Low	Test the ESP32 before integration and keep a spare board if possible.
R4	Power supply or battery failure	High	Medium	Use a fully charged, stable power source and check battery voltage regularly.
R5	LED or buzzer malfunction	Medium	Low	Test all components before deployment and replace faulty components.

R6	Software bugs in ESP32 program	High	Medium	Test the program thoroughly and debug before final demonstration.
R7	Loose wiring or poor connections	Medium	Medium	Use proper wiring, soldering, and secure all connections.
R8	Delay in project completion	Medium	Medium	Follow the project schedule and assign tasks equally among team members.